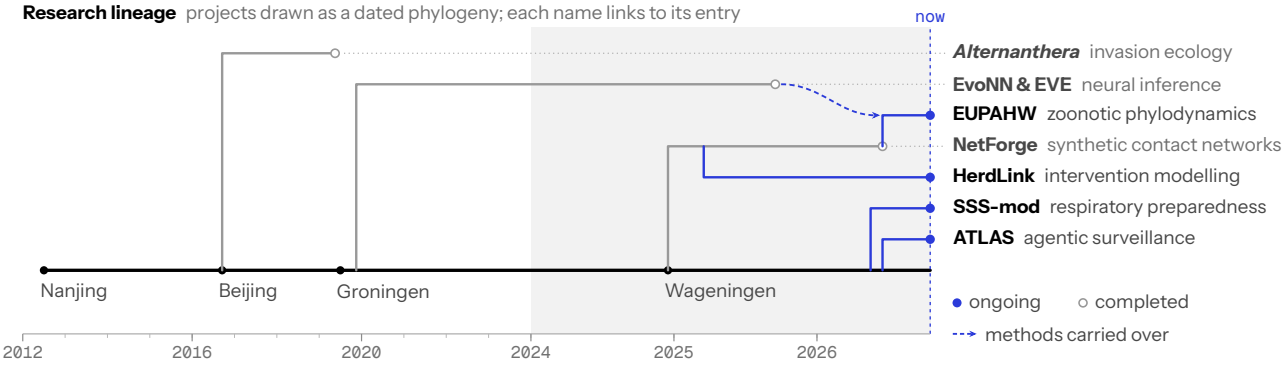
Tianjian Qin PhD

Computational biologist & AI/ML researcher
Postdoctoral researcher, Wageningen University & Research

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Life is interconnected. I build tools to understand how.



Profile Computational biologist developing machine learning methods, stochastic models and research software. My work spans neural inference from phylogenies, livestock movement networks and infectious disease modelling, with reproducible computing and interactive tools for surveillance and simulation.

Focus Computational biology, AI/ML, infectious disease modelling, network science, phylodynamics, One Health, research software.

Languages Chinese, native. English, proficient. Dutch, A2 certified with strong reading. German, A2.

Research experience

2025 – now
Postdoctoral Researcher
Wageningen University & Research, Infectious Disease Epidemiology Group
Develop computational methods and research software for One Health and pandemic preparedness. Work across livestock movement networks, hepatitis E surveillance and respiratory transmission models; build synthetic populations, reproducible analysis pipelines and interactive tools for scenario comparison.

2019 – 2026
Doctoral Researcher in Theoretical & Computational Biology
University of Groningen, Groningen Institute for Evolutionary Life Sciences
Developed simulation and inference methods for evolutionary diversification; built research packages in R, C++ and Python with CPU/GPU computing, testing, documentation and open releases.

Education

2019 – 2026
PhD in Theoretical Biology
University of Groningen, Netherlands
Thesis: *Diversification Models and Neural Inference*. Supervisors: Rampal Etienne, Luis Valente and Koen van Benthem.

2016 – 2019
MSc in Wetland Ecology
Beijing Forestry University, China
Supervisors: Hongli Li and Feihai Yu.

2012 – 2016
BSc in Marine Biology
Nanjing Normal University, China

Honours

2025 — **Denise Kirschner Best Student Paper Prize**
Journal of Theoretical Biology, inaugural prize for graduate student research

2017–2019 — **First-Class Academic Performance Award**, Beijing Forestry University

2018 — **Graduate Academic Innovation Award**, Beijing Forestry University

Current projects

Jun 2026–now

ATLAS agentic tracking and longitudinal analysis for surveillance
Developer and maintainer, Wageningen University & Research

Develop a Python workflow combining LLM extraction, scientific review and source provenance for outbreak surveillance. Build versioned evidence, longitudinal comparisons and interactive geographic, reporting-network and One Health views.

Methods Python, agentic AI, evidence review
Software `qtj.me/atlas` (live), `github.com/EvoLandEco/ATLAS`
Programme OpenAI, ChatGPT for Academic Researchers

Jun 2026–now

EUPAHW zoonotic pathogens across pigs, wild boar and humans
Researcher, Wageningen University & Research

Develop phylodynamic and network analyses linking livestock movements, wildlife and landscape data to investigate transmission pathways of zoonotic pathogens, such as hepatitis E virus and swine influenza, and inform One Health surveillance.

Methods Phylodynamics, movement networks, One Health surveillance
Programme European Partnership on Animal Health and Welfare, SOA26

May 2026–now

SSS-mod preparedness package for future respiratory outbreaks
IMPULS preparedness application, Wageningen University & Research

Develop data preparation, recalibration, model output and interface workflows supporting reuse of a setting-specific respiratory transmission model.

Methods Python and FastAPI, React and TypeScript, population synthesis
Programme ZonMw, Implementatie-impuls COVID-19 programma

Mar 2025–now

HerdLink interactive intervention modelling
Author and developer, Wageningen University & Research

Developed within IMBIT and deployed as a public web platform linking Dutch livestock trade networks, geographic views and disease simulations. Implemented dated route and regional movement restrictions, seven scenario presets, and paired baseline and intervention comparisons of epidemic trajectories, regional burden and retained trade for scenario comparison and decision support.

Methods JavaScript, React, D3.js, transmission simulation
Software `herdlink.nl` (live), `github.com/EvoLandEco/herdlink-web`
Programme IMBIT, ZonMw Kennisprogramma Pandemische Paraatheid

2026 — proposal

NextdAI integrated AI-driven early detection system
NWO KIC, Early Detection of (Emerging) Infectious Disease Outbreaks

Co-develop AI architecture and data infrastructure for heterogeneous surveillance, anomaly detection, neurosymbolic characterisation, forecasting and decision support.

Methods AI architecture, surveillance data integration, forecasting

Earlier research & software

Dec 2024–Jun 2026

NetForge synthetic livestock contact networks
Wageningen University & Research, IMBIT

Developed within the IMBIT pandemic preparedness programme, combining stochastic block models with contact history selection to generate synthetic livestock contact networks; benchmarked them against temporal reachability and simulated infection dynamics.

Methods Stochastic block models, temporal networks, epidemic simulation
Software `qtj.me/NetForge`, `github.com/EvoLandEco/NetForge`
Publications *bioRxiv* preprint (2026), `doi:10.64898/2026.09.15.751711`

Nov 2019 – Sep 2025

EvoNN & EVE neural inference for evolutionary biology
University of Groningen

Designed ensemble neural networks that learn from tree structure, branching times and summary statistics, alongside stochastic diversification models and high-performance simulators.

Methods Neural inference, stochastic simulation, R, C++ and Python
Software `github.com/EvoLandEco/EvoNN`, `github.com/EvoLandEco/evesim`
Publications *Systematic Biology* (2026), *Journal of Theoretical Biology* (2025)

Sep 2016 – May 2019

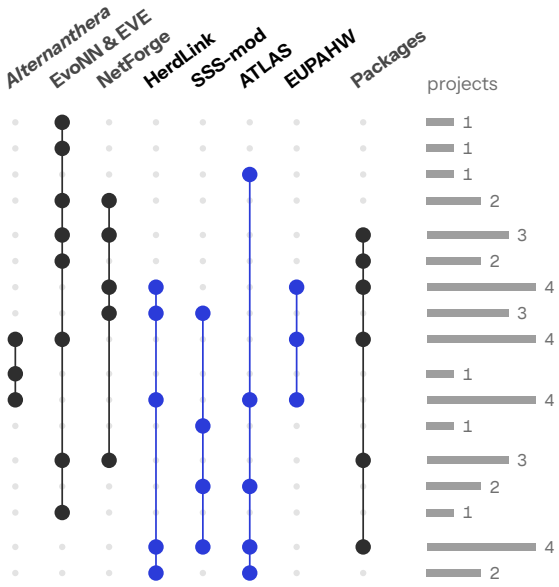
Understanding the invasion of *Alternanthera philoxeroides* in China
MSc research, Beijing Forestry University

Studied phylogenetic diversity, spatial distribution and wetland community resistance to invasion. Combined surveys across 18 Chinese provinces with controlled experiments, image analysis and phylogenetic workflows to examine community composition and environmental effects.

Methods Field surveys, controlled experiments, phylogenetic analysis
Publications *Plant Biology* (2020), *Marine and Freshwater Research* (2018)

Technical expertise

Methods by project
columns follow the research lineage; names and dots link to entries



Languages & ML	`Python` `R` `C/C++` `JavaScript` `SQL` `PyTorch` `PyTorch Geometric`
Web & graphics	`React` `D3.js` `Three.js` `Vite` `LaTeX` `TikZ` `PGFPlots`
Infrastructure	`Linux` `Bash` `Docker` `Git` `GitHub workflows`

Software contributions

Developer `NetSpectra`, `EvoLab`, `miniape`. Interactive network dynamics, lineage simulation, tree reconstruction and browser-based phylogenetic data handling.

Collaborator `treestats`, `DDD`, `DAISIE`. Phylogenetic statistics, data structures, simulation and inference for diversification and island biogeography.

Teaching, mentoring & service

Teaching Seminars and workshops at Groningen, WUR and Wageningen Bioveterinary Research on island biogeography, stochastic modelling, phylodynamics, neural networks and network statistics.

Mentoring BSc/MSc students and junior PhD researchers; pair programming on test-driven development and scientific software design.

Peer review *Systematic Biology*, *PLOS Computational Biology*, *Epidemics*, *BMC Public Health* and *BMC Plant Biology*.

Leadership Vice President, Graduate Student Council (2017–2018); student representative, Wetland Ecology Group (2016–2019), Beijing Forestry University.

Peer-reviewed articles

2026 Parameter estimation from phylogenetic trees using neural networks and ensemble learning.
Qin, T., van Benthem, K. J., Valente, L.† & Etienne, R. S.† *Systematic Biology* 75(2), 344–365
doi:10.1093/sysbio/syaf060

2025 Impact of evolutionary relatedness on species diversification and tree shape.
Qin, T., Valente, L.† & Etienne, R. S.† *Journal of Theoretical Biology* 598, 111992
doi:10.1016/j.jtbi.2024.111992

2022 Relative abundance of invasive plants more effectively explains the response of wetland communities to different invasion degrees than phylogenetic evenness.
Sun, K.-S., Liu, X.-S., **Qin, T.-J.**, Jiang, F., Cai, J.-F., Shen, Y.-L., A, S.-H. & Li, H.-L.
Journal of Plant Ecology 15(3), 625–638 doi:10.1093/jpe/rtab074

2020 Phylogenetic diversity is a better predictor of wetland community resistance to *Alternanthera philoxeroides* invasion than species richness.
Qin, T., Zhou, J., Sun, Y., Müller-Schärer, H., Luo, F., Dong, B., Li, H. & Yu, F.-H.
Plant Biology 22(4), 591–599 doi:10.1111/plb.13101

2019 Spatial environmental heterogeneity may drive functional trait variation in *Hydrocotyle vulgaris* (Araliaceae), an invasive aquatic plant.
Wan, J.-Z., Wang, M.-Z., **Qin, T.-J.**, Bu, X.-Q., Li, H.-L. & Yu, F.-H. *Aquatic Biology* 28, 149–158
doi:10.3354/ab00716

2019 Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity.
Qin, T.-J.*, Guan, Y.-T.*, Quan, H., Dong, B.-C., Luo, F.-L., Zhang, M.-X., Li, H.-L. & Yu, F.-H.
Weed Research 59(5), 377–386 doi:10.1111/wre.12373

2018 Sediment type and nitrogen deposition affect the relationship between *Alternanthera philoxeroides* and experimental wetland plant communities.
Qin, T.-J., Guan, Y.-T., Zhang, M.-X., Li, H.-L. & Yu, F.-H.
Marine and Freshwater Research 69(5), 811–822 doi:10.1071/MF17335

2018 Effects of water regime on the growth of the submerged macrophyte *Ceratophyllum demersum* at different densities.
Liu, L., Guan, Y.-T., **Qin, T.-J.**, Wang, Y.-Y., Li, H.-L. & Zhi, Y.-B.
Journal of Freshwater Ecology 33(1), 45–56 doi:10.1080/02705060.2017.1422043

Preprints

2026 From movement to spread: generating livestock contact networks that preserve infection dynamics.
Qin, T., Atamer Balkan, B., Schmid, B. V. & ten Bosch, Q. A. *bioRxiv* preprint
doi:10.64898/2026.09.15.751711

2026 Identifying evolutionary relatedness effects on diversification from phylogenies using neural networks.
Qin, T., van Benthem, K., Valente, L.† & Etienne, R. S.† *bioRxiv* preprint
doi:10.64898/2026.01.12.699140

* Equal contribution. † Joint senior authors.

Talks & workshops

Oct 2026 **Trade networks, transmission, intervention**
Led workshop, WUR MIND Day 2026, Wageningen, Netherlands.

Aug 2026 **From livestock movements to infection dynamics: mechanism-informed learning for synthetic contact networks**
Oral presentation, ModAH Conference 2026, Nantes, France. With B. Atamer Balkan, B. Schmid and Q. ten Bosch.

Aug 2018 **Phylogenetic diversity predicts invasion resistance in wetland plant communities**
Oral presentation, Fifth National Conference on Invasion Biology, Urumqi, China. With J. Zhou, Y. Guan, M. Zhang, F. Luo, B. Dong, H. Li and F. Yu.